



FORT STREET HIGH SCHOOL

YEAR 12

TRIAL HSC 2004

MATHEMATICS

*Time allowed: 3 hours
Plus Reading Time: 5 minutes*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions
- The marks allocated for each question are indicated.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board approved calculators may be used.
- Each new question is to be started on a **new page**.
- If required additional paper may be obtained from the Examination Supervisor on request.

Name: Feiyi Zhang Class: 12A

Teacher: Mr Bayas

Question No	1	2	3	4	5	6	7	8	9	10	Total
Marks	8 /12	12 /12	7 /12	10 /12	6 /12	10 /12	10 /12	5 /12	10 /12	6 /12	84/120 184

Question 1 (12 marks)

Marks:

a) Simplify $\frac{1}{2}\sqrt{48} - \sqrt{12} + \sqrt{147}$

2

b) Factorise $x^2 - 4y^2 - x - 2y$

2

c) Solve the equation $\frac{x-6}{3} - \frac{x-1}{2} = 1$

2

d) Solve $\tan x^\circ = 1$ for $0^\circ \leq x^\circ \leq 360^\circ$

1

e) Use the table of standard integrals to evaluate $\int \frac{1}{\sqrt{x^2+16}} dx$

2

f) A man earns \$91,500 in 1998 and invests 15% of his earnings in an account earning 10% p.a. compounded annually. How much interest has he accrued at the end of 5 years.

3

91500 1998

$$x^2 - 4y^2 - x - 2y$$

$$x^2 - x - 4y^2 - 2y$$

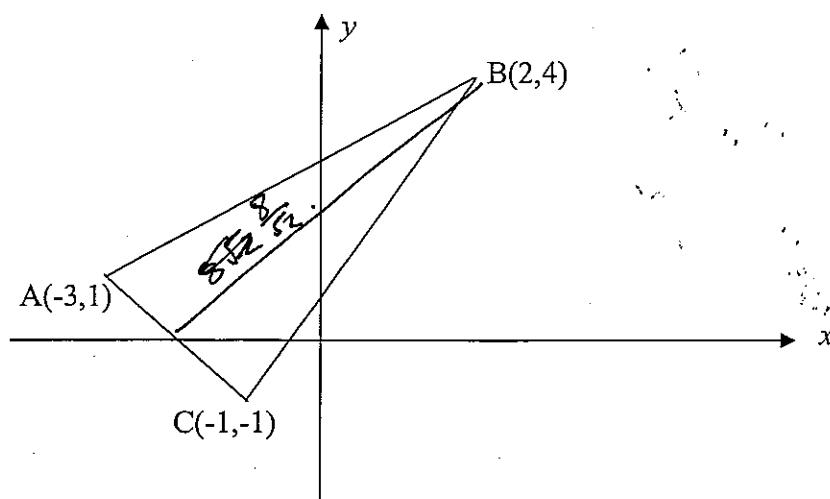
$$x^2 - x$$

$$x(x-1) - 2y(2y-1)$$

Marks:

Question 2: (12 marks)

a)



In the diagram the co-ordinates of A, B and C respectively are ^A ^B ^C $(-3,1)$, $(2,4)$ and $(-1,-1)$.

i) Show $\triangle ABC$ is isosceles.

2

ii) Find the equation of AC.

2

iii) Find the perpendicular distance from B to the line AC.

2

iv) Find the area of $\triangle ABC$.

2

b) A parabola has equation

$$(x-1)^2 = 12(y+2)$$

i) What is the vertex?

1

ii) What is the focal length?

1

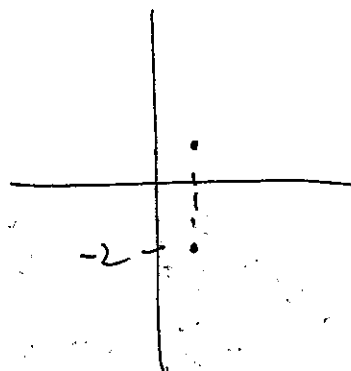
iii) Find the co-ordinates of the focus.

1

iv) Find the equation of the directrix.

1

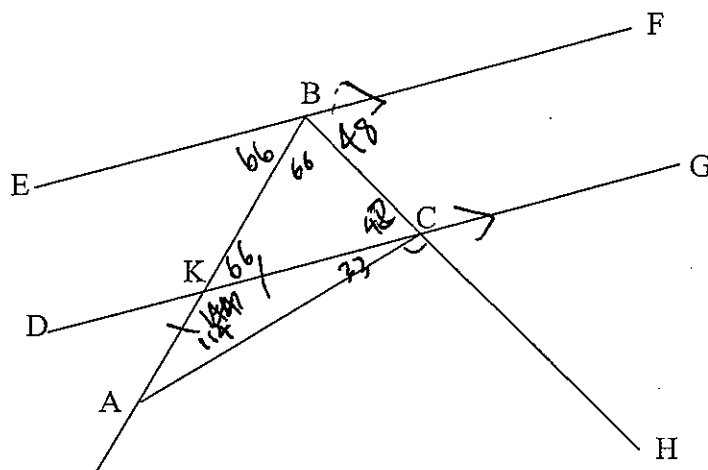
$$\frac{5}{3} \times \frac{5}{3} = \frac{25}{9}$$



Question 3: (12 marks)

Marks

a)



For the diagram above $EF \parallel DG$, $KC = KA$

$\angle EBA = 66^\circ$, $\angle FBC = 48^\circ$.

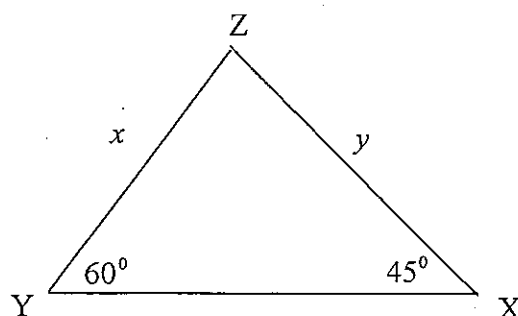
i) Copy the diagram showing the information given

1

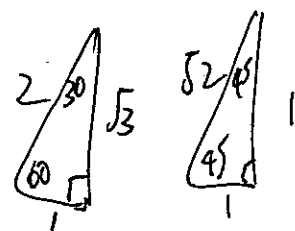
ii) Find the size of $\angle ACH$ giving reasons for your answer

3

b)



NOT TO SCALE



3

In the above diagram $\angle YXZ = 45^\circ$, $\angle XYZ = 60^\circ$. Find the exact value of the ratio $\frac{x}{y}$.

c) Jillian walks 1.2 km from point S on a bearing of 178° to point T, then turns due east and walks a further 1.6 km to point V

i) Draw a diagram representing this information, showing the size of $\angle STV$.

ii) Calculate her distance (to the nearest 10^{th} of a kilometre) and bearing (to nearest degree) from S.

1

4

$$\frac{2}{\sqrt{2}} \times \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{2} \times \sqrt{3}} = \frac{2}{\sqrt{6}}$$

$$2^{\frac{1}{2}} + 3^{\frac{1}{2}} =$$

$$\frac{\sqrt{3} \times 2}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$$\frac{2\sqrt{2}}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Marks:

Question 4: (12 marks)

a) Find the value of e^π correct to 3 significant figures

1

b) Differentiate the following functions

i) $(2x+1)^8$

2

ii) $\frac{x}{\log_e x}$

2

c) Find the following integrals

i) $\int e^{3x+1} dx$

2

ii) $\int \frac{5x-1}{5x-2} dx$

2

iii) If $\int_0^{\ln 2} \frac{e^x}{e^x+1} dx = \ln a$, find the value of a .

3

Marks:

Question 5: (12 marks)

a) The first three terms of an arithmetic series are $25 + 19 + 13 + \dots$

i) Find the 20th term

1

ii) How many terms will it take for the sum of the terms to become negative

2

b) If the sum of n terms of a series is given by the formula $S_n = 3n^2$ find an expression for the n th term and show that the series is arithmetic

3

c) For the series $2e^{-1} + 4e^{-2} + 8e^{-3} + \dots$

i) Find S_n

2

ii) Find S_∞

2

iii) Find an expression for $S_\infty - S_n$

2

$$3n^2 = \frac{n}{2} [2a + (n-1)d]$$

$$6n^2 = n(2a + (n-1)d)$$

$$6n = 2a + (n-1)d$$

$$6n = 2a + dn - d$$

$$3n = \frac{n}{2} (2a + d)$$

$$6n = n(2a + d)$$

$$T_n = a + (n-1)d$$

$$2e^{-1} + 2$$

$$4e$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_n = \frac{n(n-1)}{r-1}$$

$$S_n = \frac{a(r^n - 1)}{r-1}$$

$$T_n = a + (n-1)d$$

$$\left(\frac{2}{e^1}\right)^2 = \frac{4}{e^2}$$

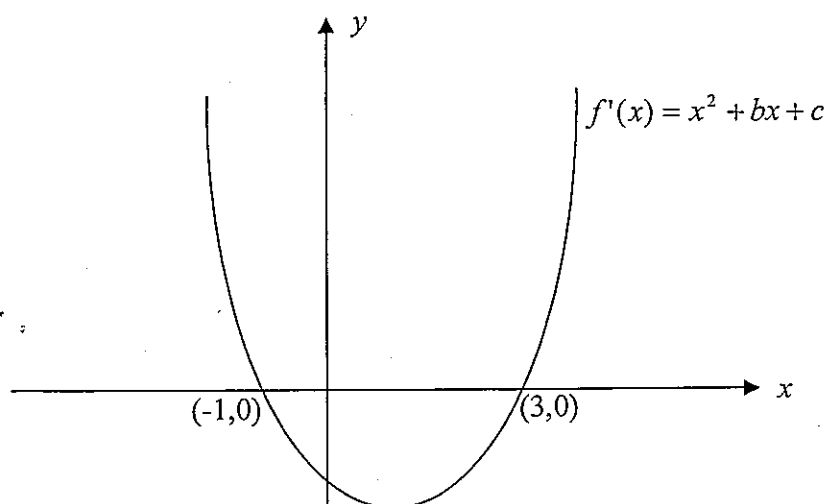
$$(4e^{-2})(2e^{-1})$$

$$2e^{-1} \left(\frac{1}{1-1}\right)$$

$$= 8e^{-3}$$

Question 6: (12 marks)

a)



$$f'(x) = x^2 + bx + c$$

$$f(x) = \frac{x^3}{3} + \frac{bx^2}{2} + cx + C$$

$\text{at } x = (-1, 0)$

$$0 = \frac{-1^3}{3} + \frac{b(-1-1)^2}{2} + C$$

The gradient function $f'(x) = x^2 + bx + c$ has been sketched in the above diagram

i) Show that $b = -2$ and $c = -3$

2

ii) State the values of x for which $y = f(x)$ has turning points and determine the nature of the turning points of $y = f(x)$

2

iii) If $y = f(x)$ passes through the point $(0, 1)$ find $f(x)$.

2

iv) Show that the point of inflection of $y = f(x)$ occurs at $\left(1, -2\frac{2}{3}\right)$.

2

b)

i) A box of 12 golf balls contains 8 white and 4 yellow balls. A boy selects 2 golf balls at random. What is the probability they are different colours?

2

ii) Another box of 12 golf balls contains 6 white, 4 yellow and 2 pink balls. A boy selects 2 golf balls at random. What is the probability they are different colours?

2

$$y = x^2$$

$$y' = 2x$$

$$y = \frac{1}{3}x^3$$

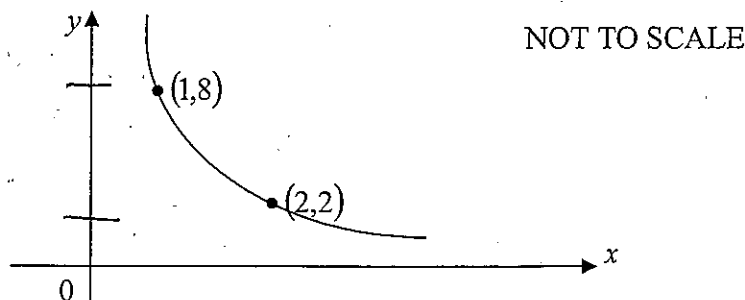
$$y' = x^2$$

$$y = \frac{1}{2}x^2$$

$$y' = x$$

Question 7: (12 marks)

a)



$$7 \int \frac{8}{y} \cdot \frac{1}{8}$$

The diagram shows the graph of $y = \frac{8}{x^2}$ (i.e. $x^2 = \frac{8}{y}$) for $x > 0$.

3

The arc of the graph between (1, 8) and (2, 2) is rotated about the y axis. Find the volume of the solid formed (in exact form).

b) Consider the function $y = 5xe^{-x}$.

i) Copy and complete the following table, giving the value to 2 decimal places.

x	0	1	2	3	4
y	0	1.84	1.35		0.37

1

ii) Find $\int_0^4 5xe^{-x} dx$ using Simpson's Rule with 5 function values.

2

iii) Find $\frac{dy}{dx}$.

2

iv) Find the value of x for which $\frac{dy}{dx} = 0$.

1

v) Show that $\frac{d^2y}{dx^2} = 5e^{-x}(x-2)$.

2

vi) What is the x coordinate of the point of inflection on the graph of $y = 5xe^{-x}$?

$$y = 2x^2 \quad 1$$

$$y' = 4x$$

Question 8: (12 marks)

Marks:

- a) For what values of k will the roots of the quadratic equation

3

$$kx^2 - 2(k+1)x + 4 = 0 \text{ be real?}$$

- b) Given that one root of the quadratic equation $3x^2 + bx + c = 0$ is three times the other root prove that $b^2 - 16c = 0$.

3

c)

- i) From a packet of mixed seed it was estimated that the probability of any seed planted yielded a white carnation was 0.02.
If n seeds are planted, write down expressions for

(α) The probability of no white carnations

1

(β) The probability of at least one white carnation.

2

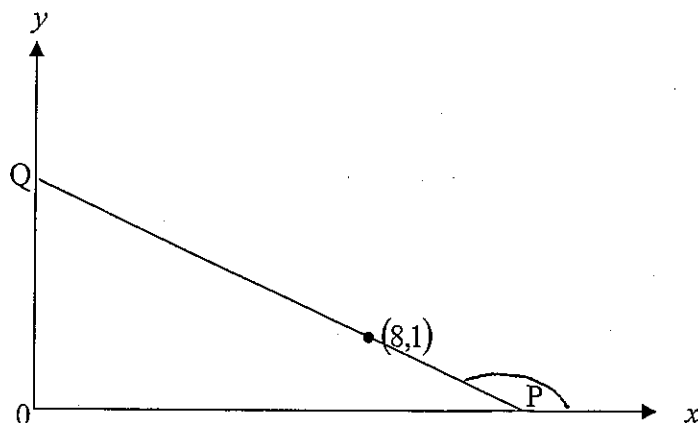
- ii) How many seeds must be planted for you to be at least 98% certain of obtaining a white carnation.

3

$$\begin{aligned} & \cancel{4(k-2)^2} \\ & (-2k-2)(-2k-2) \\ & = 4k^2 + 4k + 4k + 4 \\ & = 4k^2 + 8k + 4. \end{aligned}$$

Question 9: (12 marks)

Marks:



$$-m^{-1} = 1m^{-2}$$

A line is drawn through the point $(8,1)$ to cut the positive x axis at P and the positive y axis at Q . The gradient of PQ is m .

- Find the equation of PQ in terms of m . 1
- Show that the coordinates of P are $\left(\frac{8m-1}{m}, 0\right)$. 1
- Find the coordinates of Q . 2
- Show that the area of $\triangle OPQ$ is $\frac{1}{2}\left(16 - 64m - \frac{1}{m}\right)$. 2
- Find the value of m for which this area is a minimum, noting that m is negative. 4
- Show that the minimum area is 16 unit². 2

$$-1 \quad \frac{1}{2m^2} \quad \frac{1}{2}m^{-2}$$

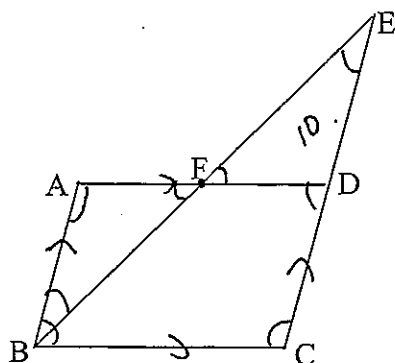
$$\frac{8m-1}{m} \times 2 = \frac{2(8m-1)}{m}$$

$$= \frac{16m-2}{m}$$

Marks:

Question 10: (12 marks)

a)



In the diagram above ABCD is a parallelogram and point F is the midpoint of AD.

- i) Show that triangles AFB and FED are congruent. 2
- ii) Show that triangles EFD and EBC are similar. 2
- iii) Hence find the area of ABCD given that the area of $\triangle EFD$ is 10 square units. 2

b)

- i) Sally has just turned 18 years old and wishes to purchase a \$25,000 car by her 25th birthday. She visits the Big Bank and discovers that she can earn interest at the rate of 0.65% per month. Assuming that it will take her exactly one month to make all the necessary financial arrangements how much must she deposit each month to achieve her goal? 3
- ii) How much less would she need to deposit monthly if her parents gave her \$3000 for her 18th birthday to use as an initial deposit? 3

END OF EXAMINATION

9 years.
96 months
- one month
= 95 months

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

TRIAL HSC SOLUTIONS

mathematics - 2004

QUESTION ONE

COMMENTS.

$$a) \frac{1}{2} \sqrt{48} - \sqrt{12} + \sqrt{147} = \frac{1}{2} \times 4\sqrt{3} - 2\sqrt{3} + 7\sqrt{3} \checkmark$$

$$= 7\sqrt{3} \quad \checkmark$$

$$b) x^2 - 4y^2 - x - 2y = (x-2y)(x+2y) - (x+2y) \checkmark$$

$$= (x+2y)(x-2y-1) \checkmark$$

$$c) \frac{x-6}{3} - \frac{x-1}{2} = 1$$

$$2(x-6) - 3(x-1) = 6 \checkmark$$

$$-x - 9 = 6$$

$$x = -15 \quad \checkmark$$

$$d) \tan x^\circ = 1$$

$$x = 45^\circ \text{ or } 225^\circ \quad \checkmark$$

$$e) \int \frac{1}{\sqrt{x^2+16}} dx = \ln(x + \sqrt{x^2+16}) \checkmark \checkmark$$

$$f) \text{ "The amount invested" } \checkmark$$

$$\text{Amount invested} = \$13725 \quad \checkmark$$

$$\text{Interest} = \$13725 \times 1.1^5 - 13725 \checkmark$$

$$= \$8379.25 \quad \checkmark$$

QUESTION TWO

COMMENTS

$$2) (i) AB = \sqrt{3^2 + 5^2} = \sqrt{34} \quad \checkmark$$

$$BC = \sqrt{5^2 + 3^2} = \sqrt{34} \quad \checkmark$$

a) i) Well Done

$\therefore \Delta ABC$ is isosceles

$$(ii) \text{ Equation of AC is } \frac{y-1}{x+3} = \frac{-1-1}{-1+3} \quad \checkmark$$

ii) Well Done

$$2y - 2 = -2x - 6$$

$$2x + 2y + 4 = 0$$

$$x + y + 2 = 0 \quad \checkmark$$

$$(iii) d = \frac{|2 + 4 + 2|}{\sqrt{1^2 + 1^2}} \quad \checkmark$$

$$= \frac{8}{\sqrt{2}} \quad \checkmark \quad (4\sqrt{2})$$

iii) many students need to learn the 'perpendicular distance' formula & to remember eqn. of line being used must be in general form so that the correct values for 'a', 'b', 'c' are substituted

$$(iv) \text{ Area} = \frac{1}{2} \times \sqrt{2^2 + 2^2} \times \frac{8}{\sqrt{2}} \quad \checkmark$$

$$= \frac{1}{2} \times 2\sqrt{2} \times \frac{8}{\sqrt{2}} \quad \checkmark$$

$$= 8 \text{ units} \quad \checkmark$$

iv) Well Done

$$b) (i) \text{ Vertex} = (1, -2) \quad \checkmark$$

b) Well Done

$$(ii) \text{ Focal length} = 3 \quad \checkmark$$

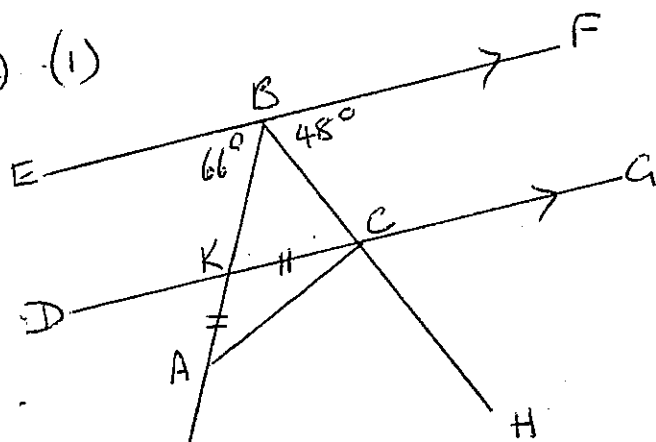
$$(iii) \text{ Focus is } (1, 1) \quad \checkmark$$

$$(iv) \text{ Directrix is } y = -5 \quad \checkmark$$

QUESTION THREE

COMMENTS

a) (i)



students made $KC = AC$.
otherwise good.

(ii) $\angle BKC = 66^\circ$ - alternate angles

$\therefore \angle CKA = 114^\circ$ - straight angle

$\angle KCA = \frac{1}{2} \times 66^\circ$ - isosceles Δ

$\angle BCK = 48^\circ$ - alternate angles

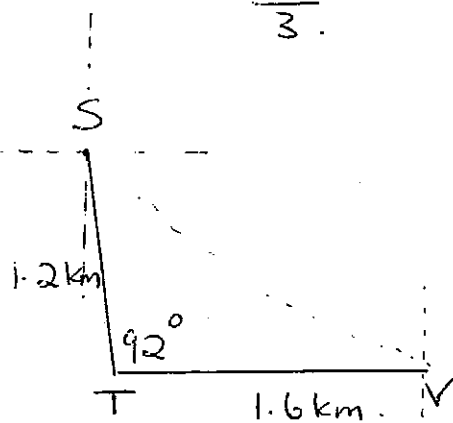
$\therefore \angle ACH = 180^\circ - (33 + 48)$
 $= 99^\circ$.

b) $\frac{x}{\sin 45^\circ} = \frac{y}{\sin 60^\circ}$ ✓

$$\begin{aligned} \frac{x}{y} &= \frac{\sin 45^\circ}{\sin 60^\circ} \\ &= \frac{1}{\sqrt{2}} \times \frac{2}{\sqrt{3}} \\ &= \frac{2}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} \\ &= \frac{\sqrt{6}}{3}. \end{aligned}$$

well done

c) (i)



many students drew
v. west of T.

QUESTION THREE (contd)

(11) $d^2 = 1.2^2 + 1.6^2 - 2 \times 1.2 \times 1.6 \cos 92^\circ$ ✓ students forgot to square root at the end.
 $d = 2.0 \text{ km to nearest } 0.1 \text{ km}$ ✓

$$\frac{\sin \angle SVT}{1.2} = \frac{\sin 92^\circ}{d}$$

$$\sin \angle SVT = \frac{1.2 \sin 92^\circ}{d} \quad \checkmark$$

$$\therefore \angle SVT = 36^\circ \text{ to nearest degree.}$$

$$\angle TSV = 52^\circ$$

$$\therefore \text{Bearing from S} = 178 - 52 \quad \checkmark \\ = 126^\circ \text{ T.}$$

QUESTION 4

COMMENTS

a) $e^\pi = 23.1$ ✓

b) (i) $\frac{d}{dx} (2x+1)^8 = 8(2x+1)^7 \times 2$ ✓
 $= 16(2x+1)^7$ ✓

correct notation
 rarely used
 3 methods
 1) this method
 ←

(ii) $\frac{d}{dx} \frac{x}{\log_e x} = \frac{\log_e x - x \times \frac{1}{x}}{(\log_e x)^2}$ ✓

or
 2) let $y =$
 $\therefore \frac{dy}{dx} =$

$= \frac{\log_e x - 1}{(\log_e x)^2}$ ✓

or
 3) let $f(x) =$
 $\therefore f'(x) =$

← $(\log_e x)^2 \neq 2 \log_e x$

c) (i) $\int e^{3x+1} dx = \frac{1}{3} e^{3x+1} + C$ ✓✓

* constant often left out.

(ii) $\int \frac{dx}{5x-2} = \frac{1}{5} \ln(5x-2) + C$ ✓✓

(iii) $\int_0^{\ln 2} \frac{e^x}{e^x+1} dx = \ln(e^x+1) \Big|_0^{\ln 2}$ ✓

$= \ln(3) - \ln 2$ ✓

$= \ln 1.5$

$\therefore a = 1.5$ ✓

* some used a
 calculator
 rather than
 recognise $e^{\ln 2} = 2$
 nor $\ln 3 - \ln 2$
 $= \ln\left(\frac{3}{2}\right)$
 $= 1.5$

QUESTION 5

COMMENTS

a) i) $T_{20} = 25 + 19x - 6$

① $= -89$ ✓

Some - incorrect
n value, others
missed -6

(ii) $S_n = \frac{n}{2} (50 + 6(n-1))$

② $= 25n - 3n^2 + 3n$ ~~was~~

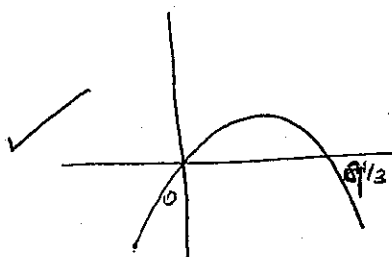
Many used
 T_n not S_n .

If $28n - 3n^2 < 0$

$n(28 - 3n) < 0$

$\therefore n > 9\frac{1}{3}$

$\therefore 10$ terms are needed for
negative sum. ✓



b) $T_n = S_n - S_{n-1}$

③ $= 3n^2 - 3(n-1)^2$ ✓

$= 3n^2 - 3[n^2 - 2n + 1]$

$= 6n - 3$ ✓

Series is 3, 9, 15, ... ✓
ie AP with $d=6$.

Few could work
out S_1, S_2 etc

Some didn't tidy
up T_n i.e.
⑦ $T_n = 3 + (n-1)6$

c) $S_n = \frac{2e^{-1}((2e^{-1})^n - 1)}{2e^{-1} - 1}$ ✓

②

$= \frac{\frac{2}{e} \left(\frac{2^n}{e^n} - 1 \right)}{\frac{2}{e} - 1}$

$= \frac{\frac{2}{e} - 1}{\frac{2}{e} \left(\frac{2^n}{e^n} - 1 \right)}$

$= \frac{2 - e}{e}$

$(2e^{-1})^n$ not
 $2e^{-n}$

$r > 1$ missed

1 for substitution

1 for tidy up

QUESTION 5

COMMENTS

$$= \frac{2}{2-e} \left[\left(\frac{2}{e} \right)^n - 1 \right] \checkmark$$

$$S_{\infty} = \frac{\frac{2}{e}}{1 - \frac{2}{e}} \checkmark$$

$$= \frac{2 \times e}{e - 2}$$

$$= \frac{2}{e-2} \checkmark$$

Some approx.
on calculator.

$$S_{\infty} - S_n = \frac{2}{e-2} - \frac{2}{2-e} \left[\left(\frac{2}{e} \right)^n - 1 \right] \checkmark$$

1 for
subtraction
of S_n & S_{∞}
from working

$$= \frac{2}{e-2} \left[1 + \left(\frac{2}{e} \right)^n - 1 \right]$$

$$= \frac{2}{e-2} \left(\frac{2}{e} \right)^n \checkmark$$

1 for correct
answer.

$$S_n = \frac{a(1-r^n)}{1-r}, \quad r < 1$$

$$= \frac{2e^{-1} [1 - (2e^{-1})^n]}{1 - 2e^{-1}}$$

$$= \frac{\frac{2}{e} [1 - 2^n e^{-n}]}{1 - \frac{2}{e}}$$

$$= \frac{2}{e} [1 - 2^n e^{-n}] \times \frac{e}{e-2} = \frac{2}{e-2} \left[1 - \left(\frac{2}{e} \right)^n \right]$$

QUESTION 6

COMMENTS

a) (i) $f'(x) = x^2 + bx + c$

$f'(3) = 9 + 3b + c = 0$ ① ✓

$f'(-1) = 1 - b + c = 0$ ② ✓

① - ② $\Rightarrow 8 + 4b = 0$

$b = -2$

$1 + 2 + c = 0$

$c = -3$ ✓

(i) $f(x)$ has turning points when

$f'(x) = 0$ if $x = -1$ and 3 .

At $x = -1$ max. turning pt. ✓

$x = 3$ min. turning pt. ✓

(iii) $f(x) = \int x^2 - 2x - 3$

$= \frac{x^3}{3} - x^2 - 3x + c$ ✓

If passes through $(0, 1)$ then

$1 = c$

$y = \frac{x^3}{3} - x^2 - 3x + 1$ ✓

(iv) $f''(x) = 2x - 2$

For point of inflection ✓

$f''(x) = 0$ and changes sign

$\therefore x = 1$ $x = 1 - \epsilon < 0$ $x = 1 + \epsilon > 0$

$\therefore$ Pt of inflection at $(1, -2\frac{2}{3})$ ✓

One mark solving
 $f''(x) = 2x - 2 = 0$

One mark finding
 $f(1)$ or

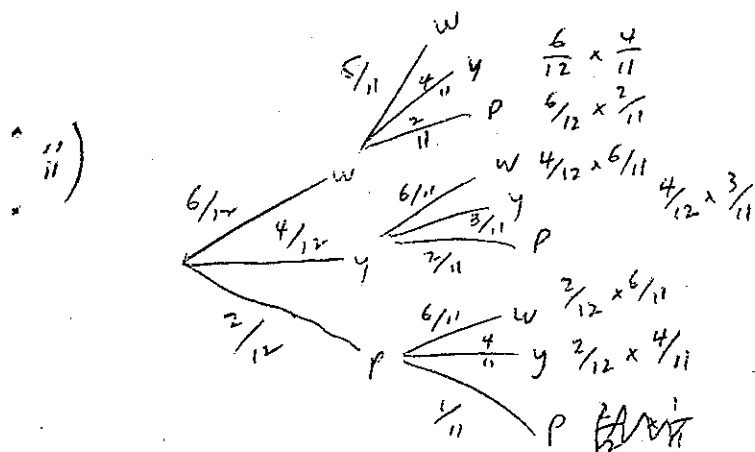
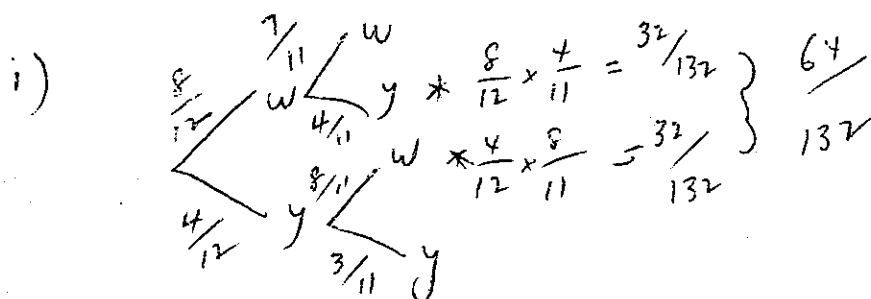
Testing for change in
concavity

Q6 contd

COMMENTS

$$\begin{aligned} \circ \text{ (i) } P(\text{both different}) &= P(WY) \text{ or } P(YW) \\ \frac{32}{432} + \frac{32}{132} &= \frac{8}{12} \times \frac{4}{11} + \frac{4}{12} \times \frac{8}{11} \checkmark \\ &= \frac{16}{33} \checkmark \end{aligned}$$

$$\begin{aligned} \cdot \text{ (ii) } P(\text{both different}) &= P(W\bar{W}) + P(Y\bar{Y}) + P(P\bar{P}) \\ &= \frac{6}{12} \times \frac{6}{11} + \frac{4}{12} \times \frac{8}{11} + \frac{2}{12} \times \frac{10}{11} \checkmark \\ &= \frac{2}{3} \checkmark \end{aligned}$$



QUESTION 7.

COMMENTS

$$\begin{aligned}
 \text{a) } V &= \pi \int_2^8 x^2 dy \\
 &= \pi \int_2^8 \frac{8}{y} dy \quad \checkmark \\
 &= \pi [8 \ln y]_2^8 \quad \checkmark \\
 &= \pi (8 \ln 8 - 8 \ln 2) \quad \checkmark \\
 &= 8\pi \ln 4 \text{ cubic units.}
 \end{aligned}$$

Well done.
although many
used wrong
limits.

$$\text{b) (i) Correct value} = 0.75 \quad \checkmark$$

$$\begin{aligned}
 \text{(ii) } \int_0^4 5xe^{-x} &= \frac{2}{6} (0 + 4 \times 1.84 + 1.35) \quad \checkmark \\
 &\quad + \frac{2}{6} (1.35 + 4 \times 0.75 + 0.37) \\
 &= 4.48 \text{ m}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) } \frac{dy}{dx} &= -5xe^{-x} + e^{-x} \times 5 \quad \checkmark \\
 &= 5e^{-x}(1-x) \quad \checkmark
 \end{aligned}$$

$$\text{(iv) } \frac{dy}{dx} = 0 \text{ if } x = 1 \quad \checkmark$$

$$\begin{aligned}
 \text{(v) } \frac{d^2y}{dx^2} &= 5e^{-x} \times -1 + (1-x) \times -5e^{-x} \\
 &= -5e^{-x} - 5e^{-x}(1-x) \quad \checkmark \\
 &= 5e^{-x}(-1-1+x) \\
 &= 5e^{-x}(x-2) \quad \checkmark
 \end{aligned}$$

$$\text{(vi) For point of inflection } x = 2 \quad \checkmark$$

QUESTION 8

Comments

a) For real roots $\Delta \geq 0$.

$$\Delta = [-2(k+1)]^2 - 4 \times k \times 4 \checkmark$$

$$= 4(k^2 + 2k + 1) - 16k.$$

$$= 4k^2 - 8k + 4 \checkmark$$

$$= 4(k^2 - 2k + 1) \checkmark = 4(k-1)^2 \checkmark$$

$$\geq 0 \text{ for all values of } k.$$

b) Roots are $\alpha, 3\alpha$

$$4\alpha = -\frac{b}{3} \quad \alpha = -\frac{b}{12}$$

$$3\alpha^2 = \frac{c}{3} \checkmark$$

$$3\left(-\frac{b}{12}\right)^2 = \frac{c}{3} \checkmark$$

$$\frac{9b^2}{144} = c.$$

$$9b^2 = 144c.$$

$$b^2 - 16c = 0. \checkmark$$

c) (i) $P(\text{no white carnations}) = (0.98)^n \checkmark$

(ii) $P(\text{at least 1 white}) = 1 - P(\text{no whites}) \checkmark$

$$= 1 - 0.98^n \checkmark$$

(iii) $1 - 0.98^n \geq 0.98 \checkmark$

$$-0.98^n \geq -0.02.$$

$$0.98^n \leq 0.02$$

$$n \ln 0.98 \leq \ln 0.02 \checkmark$$

$$n \geq \frac{\ln 0.02}{\ln 0.98} \quad n \geq 194 \checkmark$$

QUESTION 19

Comments.

2) Equation of PO is

$$(i) \quad y - 1 = m(x - 8) \quad \checkmark$$

$$y = mx - 8m + 1. \quad \checkmark$$

(ii) At P $y = 0$.

$$0 = mx - 8m + 1$$

$$mx = 8m - 1$$

$$x = \frac{8m - 1}{m}$$

$$\therefore P = \left(\frac{8m - 1}{m}, 0 \right). \quad \checkmark$$

students forgot to express these as points.

(iii) At Q, $x = 0$. $\checkmark$

$$\therefore y = 1 - 8m.$$

$$Q = (0, 1 - 8m). \quad \checkmark$$

$$(iv) \text{ Area } \Delta OPQ = \frac{1}{2} \times \left(\frac{8m - 1}{m} \right) \times (1 - 8m) \quad \checkmark$$

$$= \frac{1}{2} \left(\frac{8m - 64m^2 - 1 + 8m}{m} \right)$$

well done.

$$= \frac{1}{2} \left(\frac{16m - 64m^2 - 1}{m} \right)$$

$$= \frac{1}{2} \left(16 - 64m - \frac{1}{m} \right) \quad \checkmark$$

$$(v) \quad \frac{dA}{dm} = \frac{1}{2} \left(-64 + \frac{1}{m^2} \right) \quad \checkmark$$

$$= 0 \text{ if } m = \pm \frac{1}{8} \quad \checkmark$$

$m < 0$ since line has negative slope

* when differentiating
 $\frac{1}{2m^2}$
 $(2m)^{-2}$
 $-2(2m)^{-3} \quad ?$

$$\frac{d^2 A}{dm^2} = -\frac{1}{m^3} \quad \checkmark$$

> 0 if $m = -\frac{1}{8}$ $\therefore$ Minimum area
at $m = -\frac{1}{8}$.

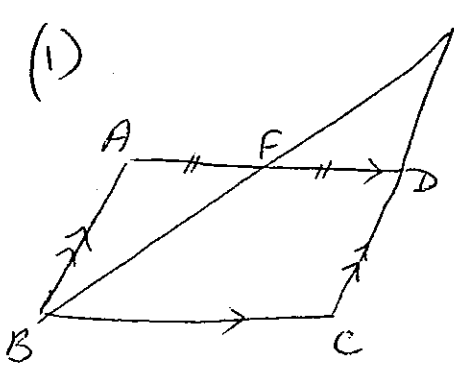
$$\begin{aligned} \text{(vi) } \text{Min. Area} &= \frac{1}{2} \left(16 - 64 \times -\frac{1}{8} - \frac{1}{-\frac{1}{8}} \right) \checkmark \\ &= \frac{1}{2} (16 + 8 + 8) \checkmark \\ &= 16 \text{ units}^2 \end{aligned}$$

$\checkmark$ - good.

QUESTION 10

COMMENT

a) (i)



In $\triangle AFB$ & $\triangle FED$

(i) $AF = FD$ given

(ii) $\angle AFB = \angle FED$ vertically opposite

(iii) $\angle ABF = \angle FED$ alternate angles ✓

$AB \parallel CE$

$\therefore \triangle AFB \equiv \triangle FED$ AAS Test ✓

(ii) In $\triangle EFD$ & $\triangle EBC$

i) $\angle E$ is common ✓

(ii) $\angle EFD = \angle EBC$ - corresponding angles ✓

$\therefore \triangle EFD \sim \triangle EBC$ since equiangular ✓

(iii) If area $\triangle EFD = 10u^2$ then

area $\triangle EBC = 4 \times 10u^2$ since ratio of sides is 1:2 ✓

$\therefore$ Since $\triangle AFB \equiv \triangle FED$ then

area ABCD is $40u^2$. ✓

$$b) 25000 = m \left[(1.0065)^{83} + (1.0065)^{82} + (1.0065)^{81} + \dots + \dots - (1.0065)^1 \right] \quad \checkmark$$

$$= m \left(1.0065 \left[\frac{(1.0065)^{83} - 1}{0.0065} \right] \right) \quad \checkmark$$

$$\therefore m = \$226.71 \text{ to nearest cent.} \quad \checkmark$$

QUESTION 10 (contd)

COMMENT

$$(ii) 25000 = 3000(1.0065)^{83} +$$

$$m \left(\frac{1.0065 \left[(1.0065)^{83} - 1 \right]}{0.0065} \right)$$

$$m = \$180.13 \text{ to nearest cent.}$$

Sally would save \$46.58 per month.

